Python DA Assignment 1

Data Analysis using NumPy and Pandas

1. Aim

To learn basic data analysis using NumPy and Pandas in Python.

2. Software Used

- Python
- Jupyter Notebook
- NumPy
- Pandas

3. NumPy

- Created a temperature array.
- Checked array shape and data type.
- Converted Celsius into Fahrenheit.
- Found maximum, minimum and average temperature.
- Used slicing to select values.
- Created a 2D array.

4. Pandas Series

- Created a Series using student marks.
- Selected values using index.
- Changed a mark.
- Deleted a value.
- Calculated CGPA.

5. Pandas DataFrame

- Created a transaction table.
- Viewed the first and last rows.
- Checked rows, columns and data types.
- Selected and filtered data.
- Grouped data by region.

- Updated and deleted records.

Result

I successfully completed basic data analysis using NumPy and Pandas in Python.